Complete Linkage (Hierarchical Agglomerative Clustering)

EXAMPLE

- Find the clusters using complete link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

EXAMPLE COMPLETE LINK

- The distance matrix is

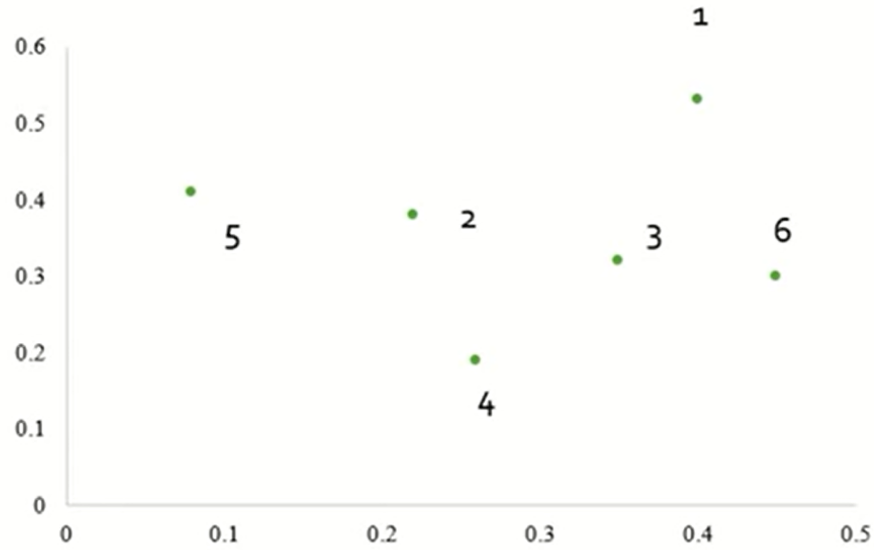
	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK



EXAMPLE COMPLETE LINK

- The distance matrix for cluster P3,P6

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P3,P6),P1)]
- MAX(dist(P3,P1), (P6,P1))

  = MAX[(0.22,0.23)]

  = 0.23
- To update the distance matrix MAX[dist(P3,P6),P2)]
- MAX(dist(P3,P2), (P6,P2))

  = MAX[(0.15,0.25)]

  = 0.25

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P3,P6),P4)]
- MAX(dist(P3,P4), (P6,P4))

  = MAX[(0.15,0.22)]

  = 0.22
- To update the distance matrix MAX[dist(P3,P6),P5)]
- MAX(dist(P3,P5), (P6,P5))

  = MAX[(0.28,0.39)]

  = 0.39

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

Example complete link

- The updated distance matrix for cluster P3,P6

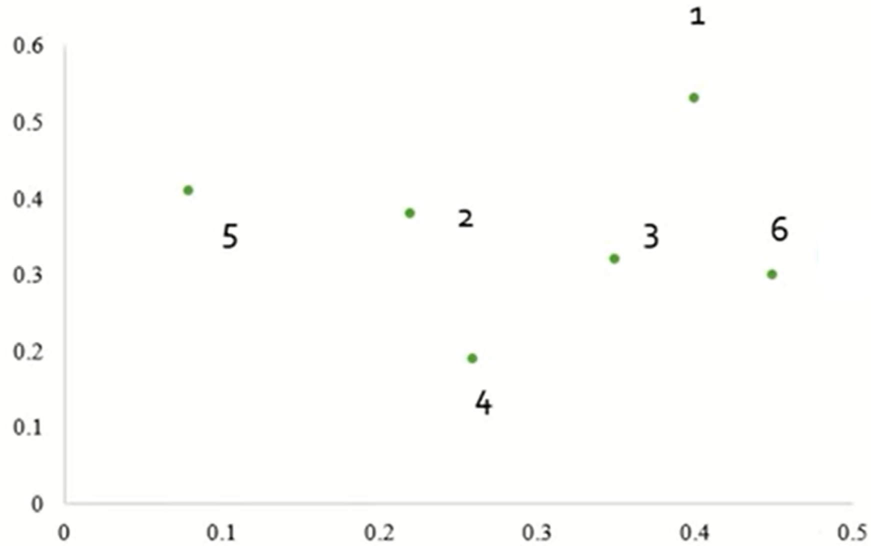
	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

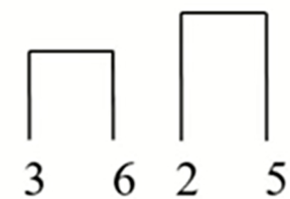
EXAMPLE COMPLETE LINK

- The distance matrix

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK





Example complete link

- The distance matrix , cluster (P2,P5)

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK

- **T**o update the distance matrix MAX[dist(P2,P5),P1)]
- MAX[dist(P2,P1), (P5,P1)]

  = MAX[(0.23,0.34)]

  = 0.34
- **T**o update the distance matrix MAX[dist(P2,P5),(P3,P6)]
- MAX[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = MAX[(0.25,0.39)]

  = 0.39

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P2,P5),P4)]
- MAX[dist(P2,P4), (P5,P4)]

  = MAX[(0.20,0.29)]

  = 0.29

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

Example complete link

- The updated distance matrix for cluster (P2,P5)

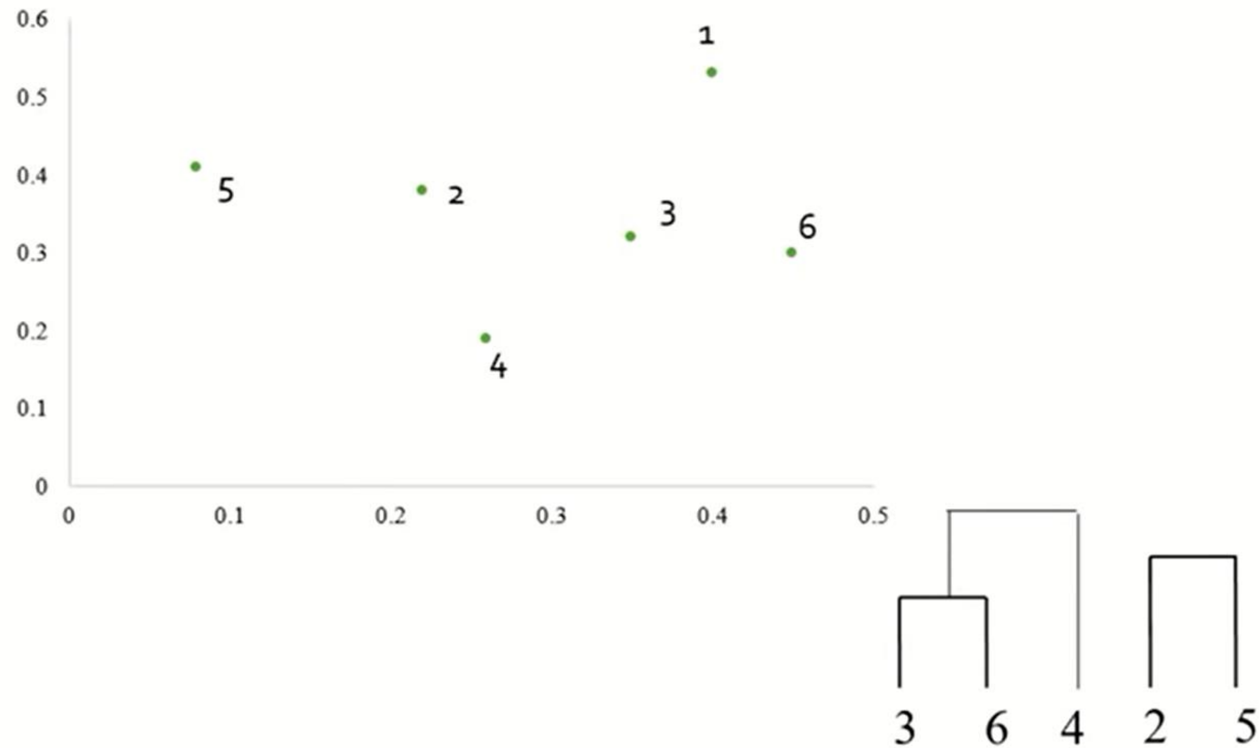
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

Example complete link

- The distance matrix is

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

EXAMPLE COMPLETE LINK



EXAMPLE COMPLETE LINK

- To update the distance matrix[((P3,P6),P4) , P1]
- MAX[dist(P3,P6),P1), (P4,P1)]

  = MAX[(0.23,0.37)]

  = 0.37
- To update the distance matrix MAX[dist((P3,P6),P4),(P2,P5)]
- MAX[dist((P3,P6),(P2,P5)), (P4,(P2,P5))]

  = MAX[(0.39,0.29)]

  = 0.39

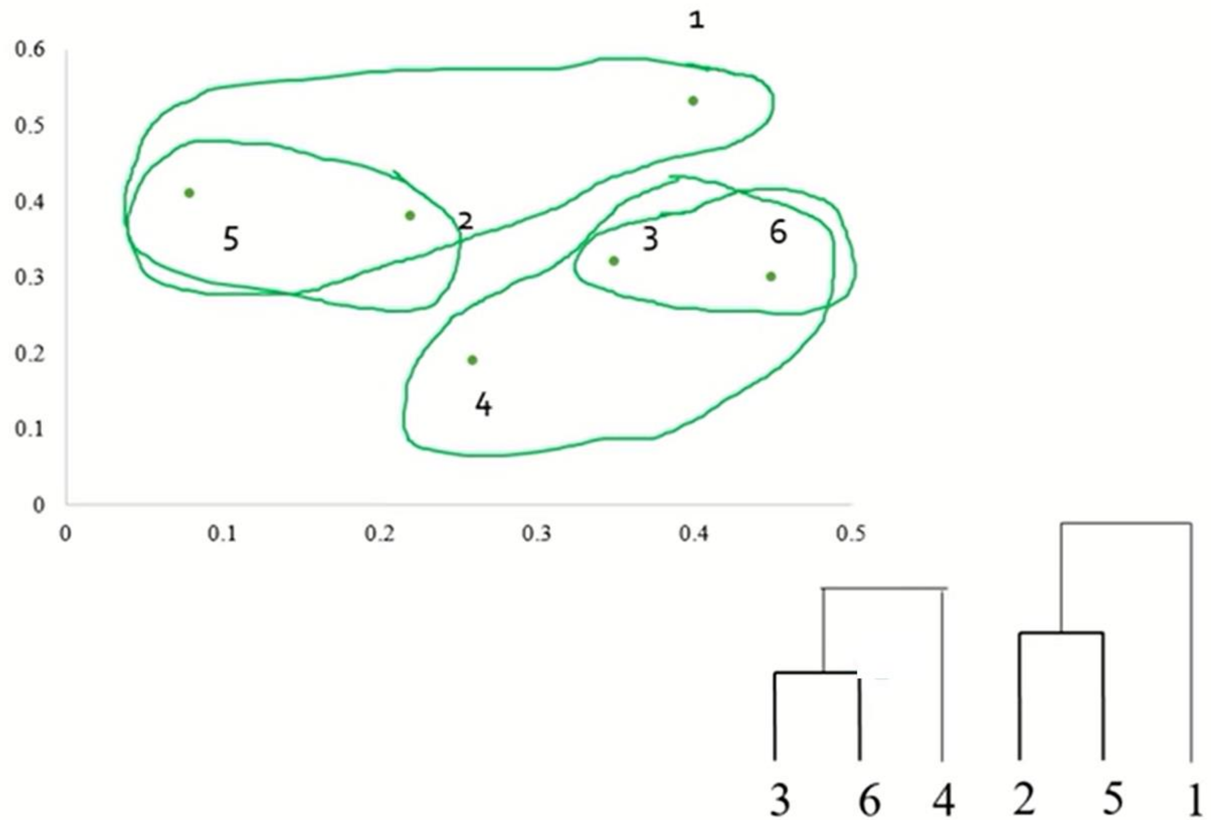
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

Example complete link

- The updated distance matrix for cluster P3,P6,P4

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.34	0	
P3,P6,P4	0.37	0.39	0

EXAMPLE COMPLETE LINK



Example complete link

- **Cluster** [(P2,P5),P1]
- MAX[dist(P2,P5),(P3,P6,P4), (P1,(P3,P6,P4))]

  = MAX[(0.39,0.37)]

  = 0.39

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.34	0	
P3,P6,P4	0.37	0.39	0

EXAMPLE COMPLETE LINK

- The updated distance matrix for cluster P2,P5,P1

	P2,P5,P1	P3,P6,P4
P2,P5,P1	0	
P3,P6,P4	0.39	0

EXAMPLE COMPLETE LINK